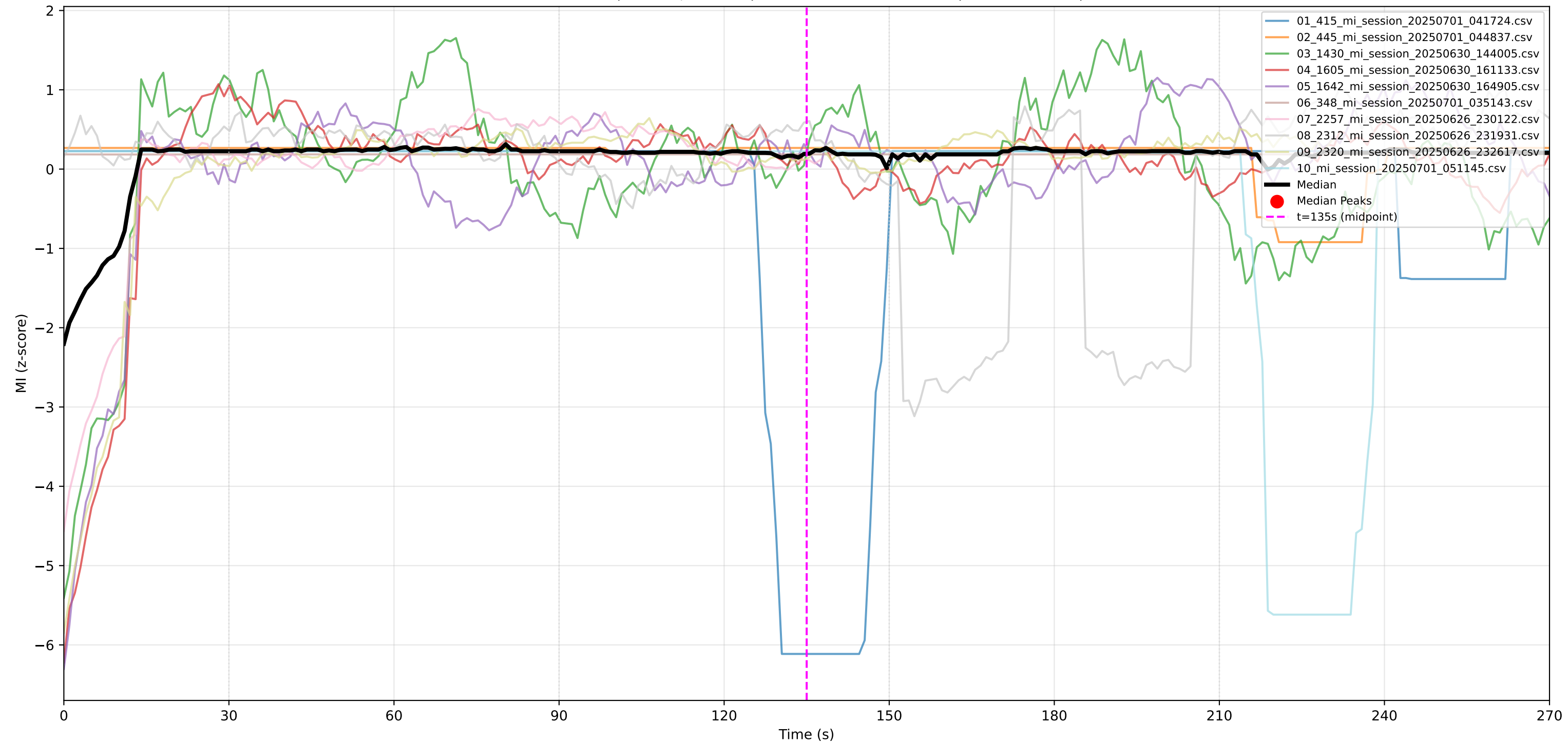
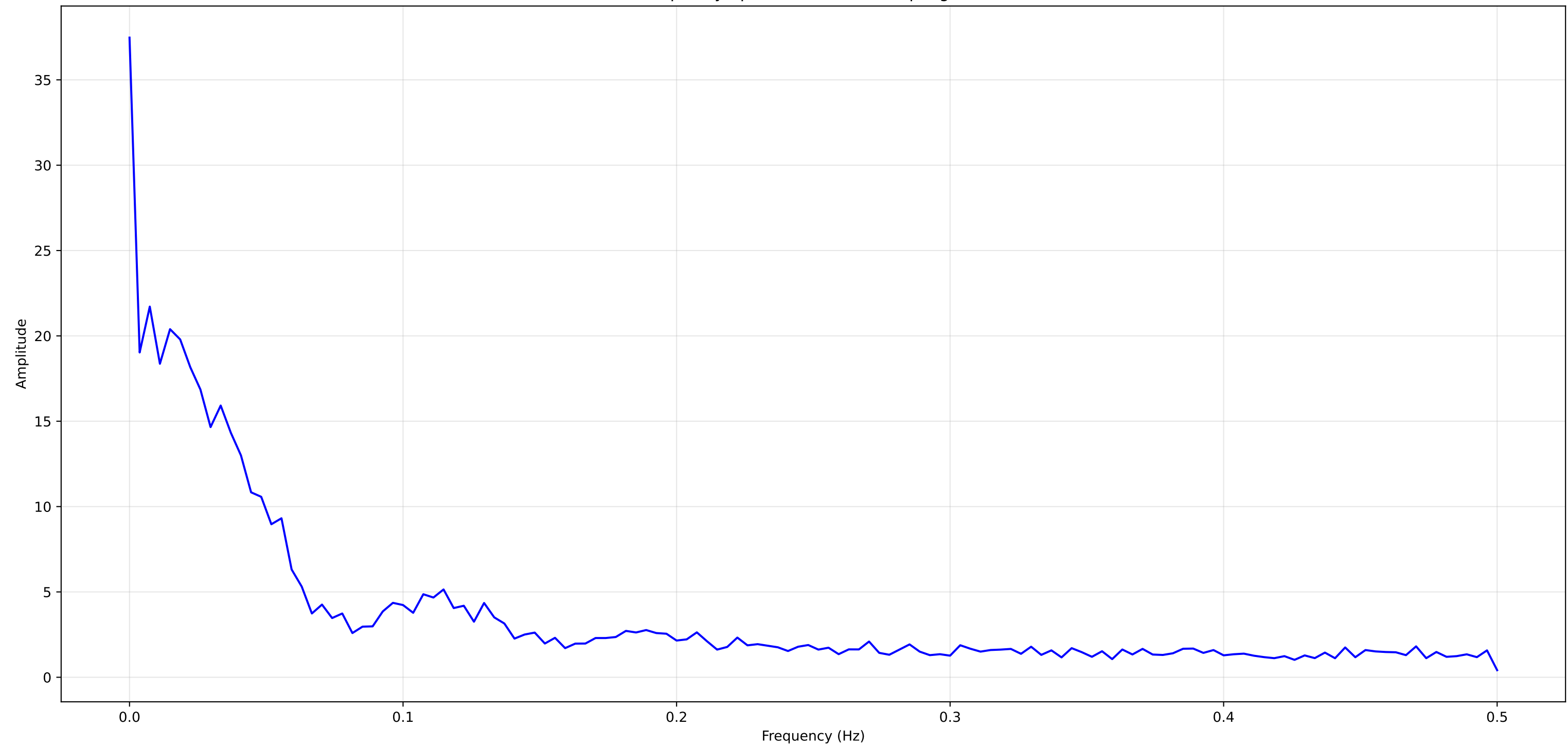


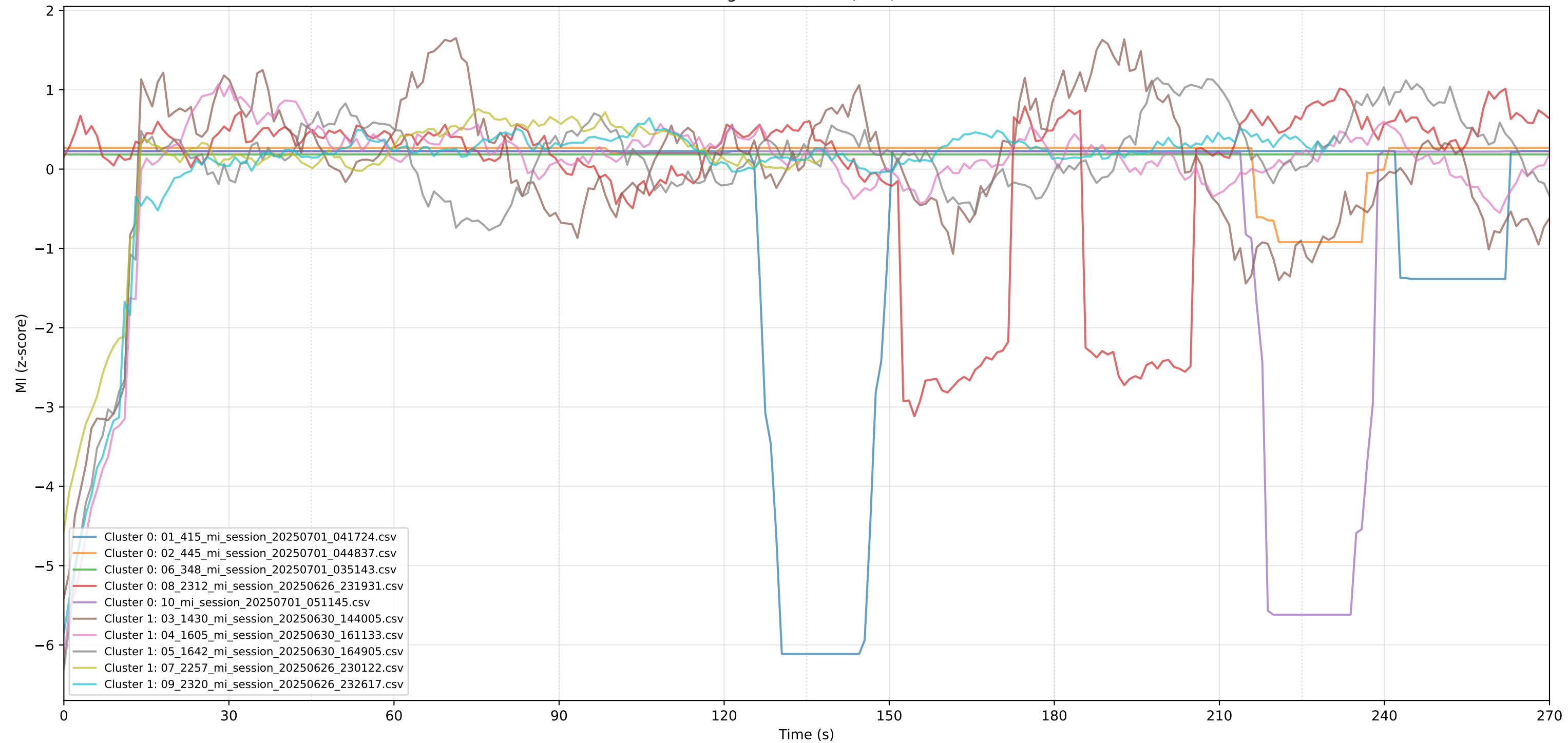
MI Curves (z-score, MA=20) with Median and Peaks (270s Duration)



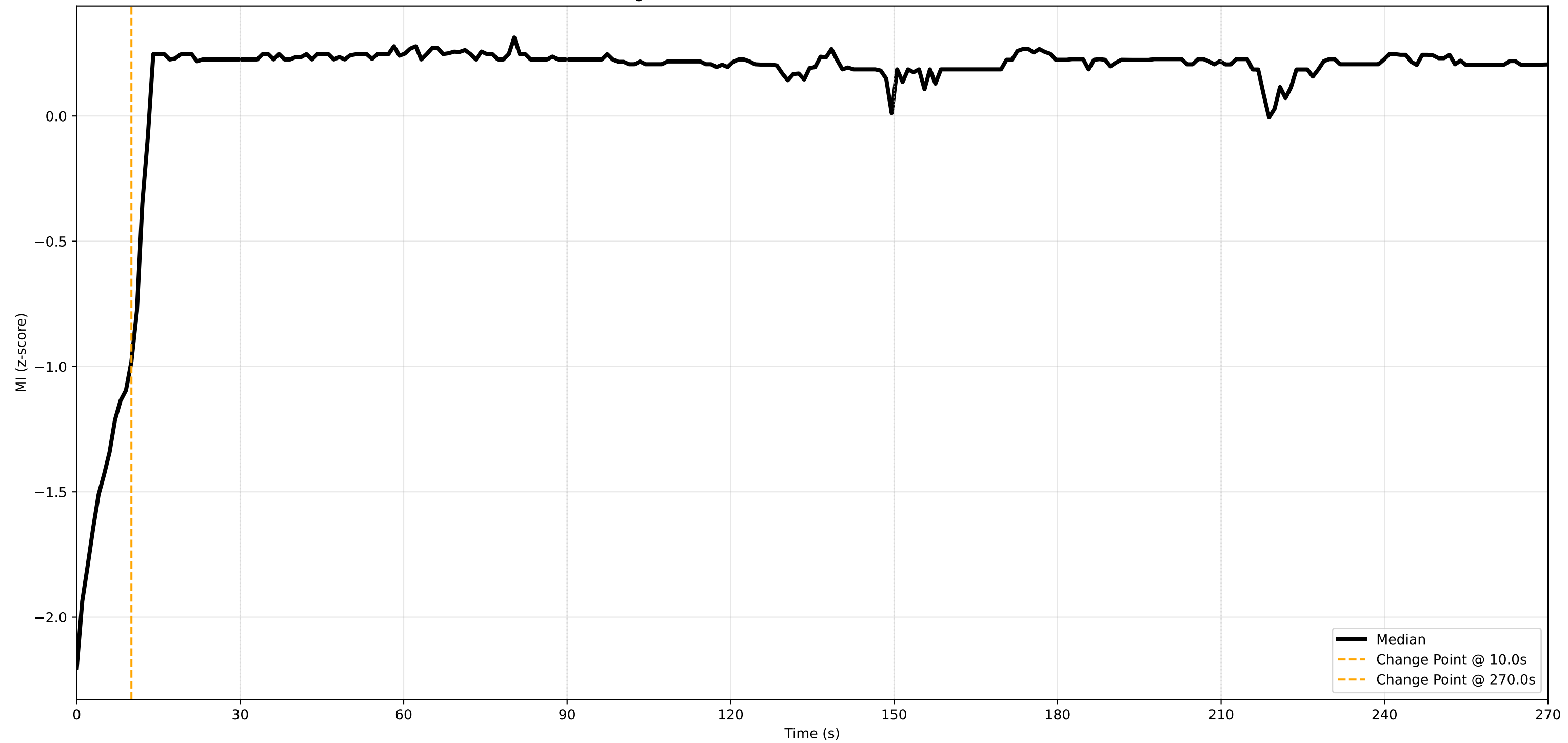
Median MI Frequency Spectrum (FFT) - Sampling Rate: 1.000 Hz



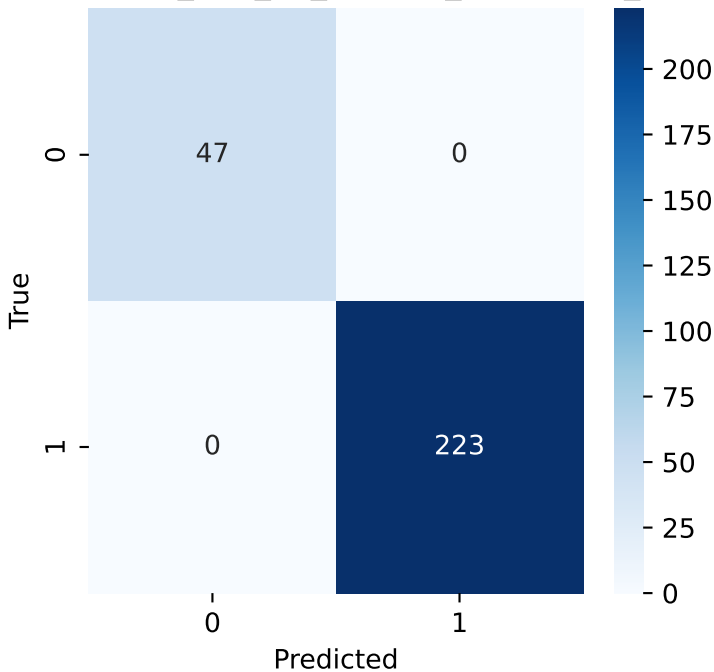
KMeans Clustering of MI Curves (k=2) - 270s Duration



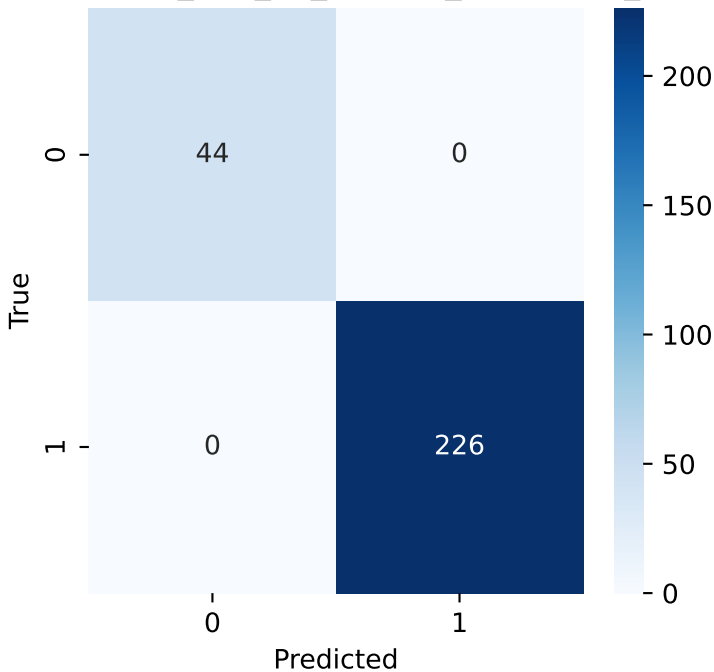
Change Point Detection on Median MI - 270s Duration



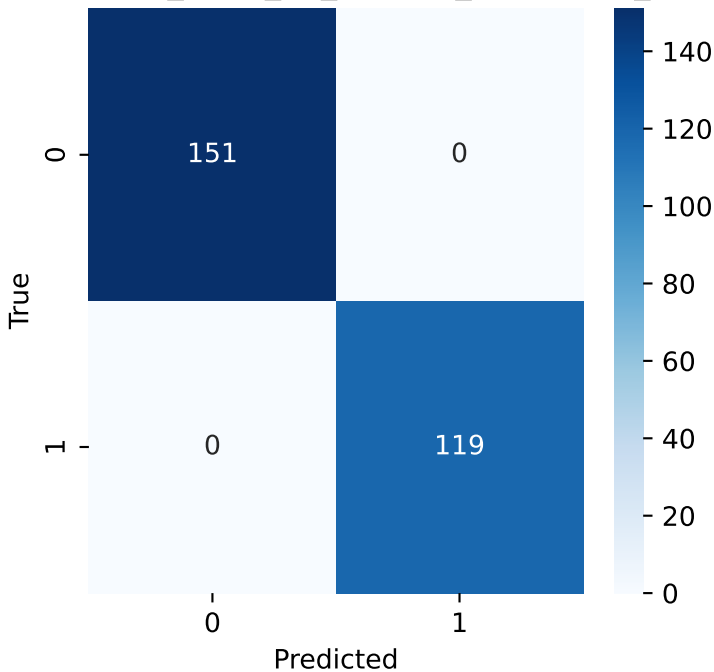
on Matrix: 01_415_mi_session_20250701_04172



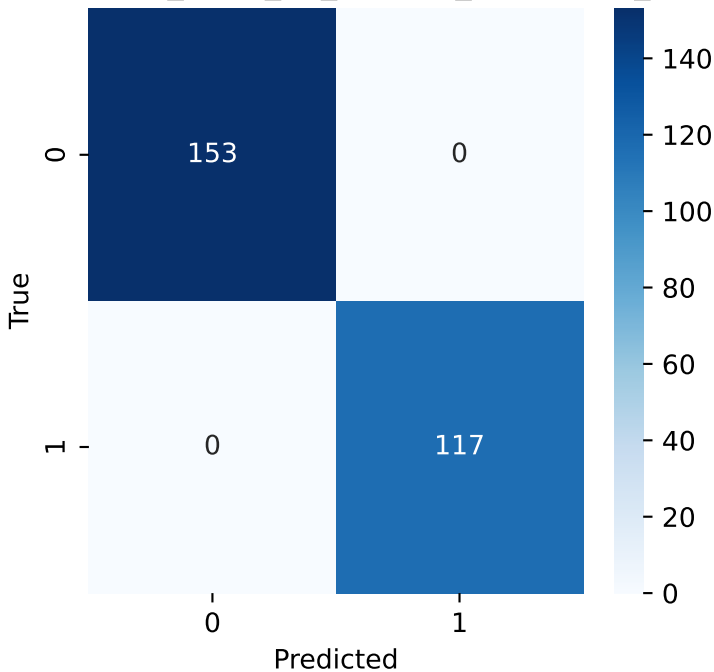
on Matrix: 02_445_mi_session_20250701_04483



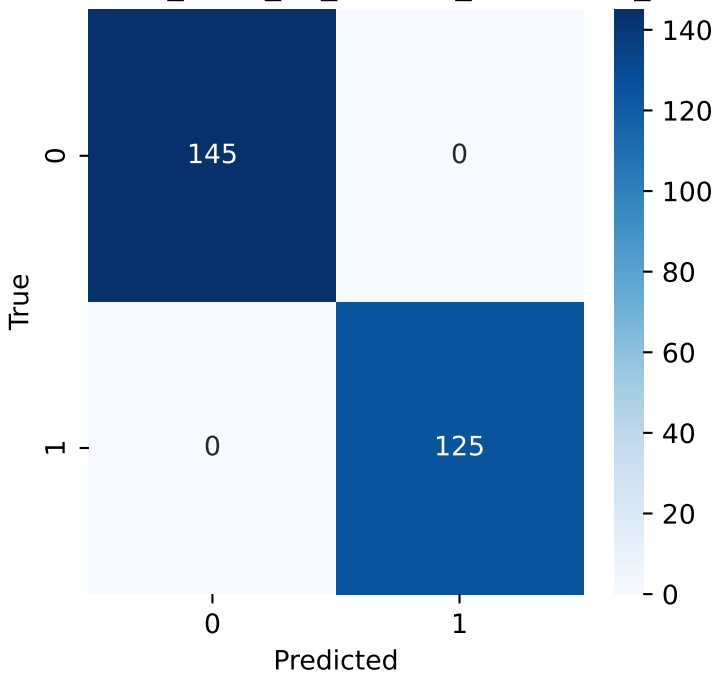
n Matrix: 03_1430_mi_session_20250630_14400



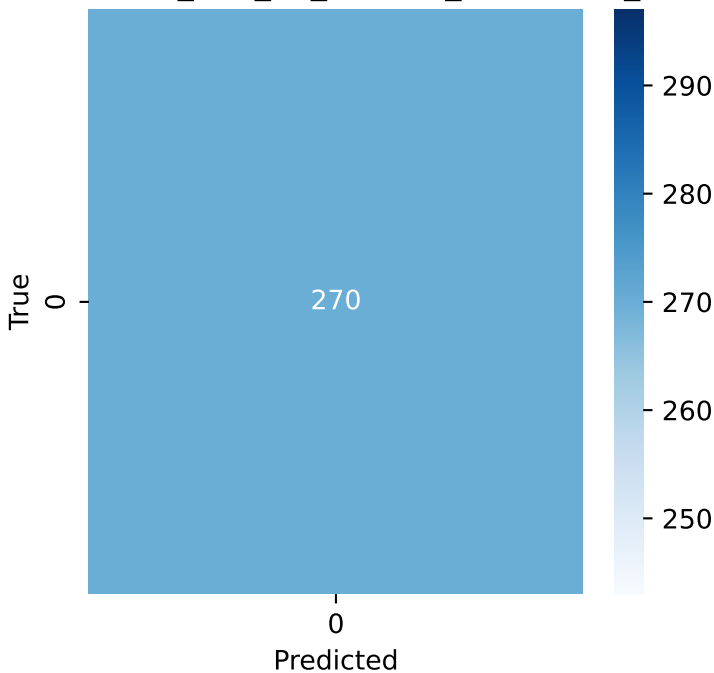
n Matrix: 04_1605_mi_session_20250630_16113



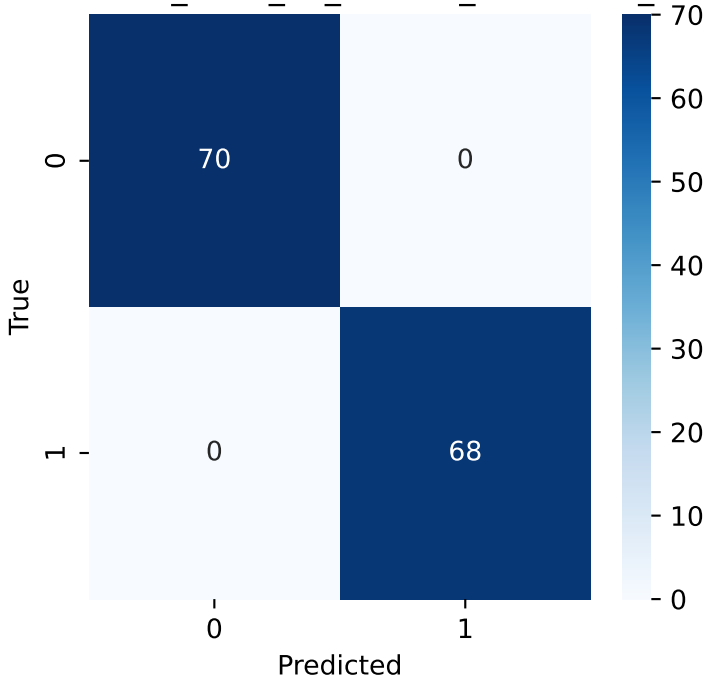
n Matrix: 05_1642_mi_session_20250630_16490



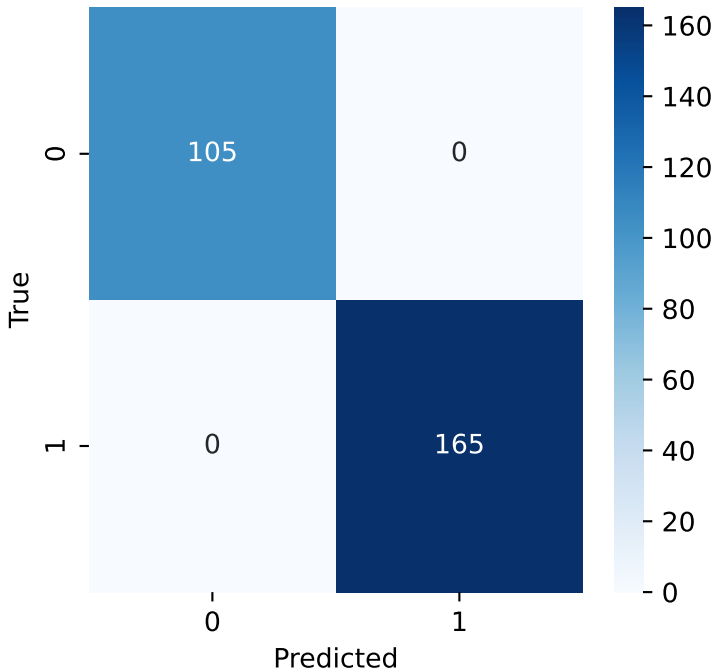
on Matrix: 06_348_mi_session_20250701_03514



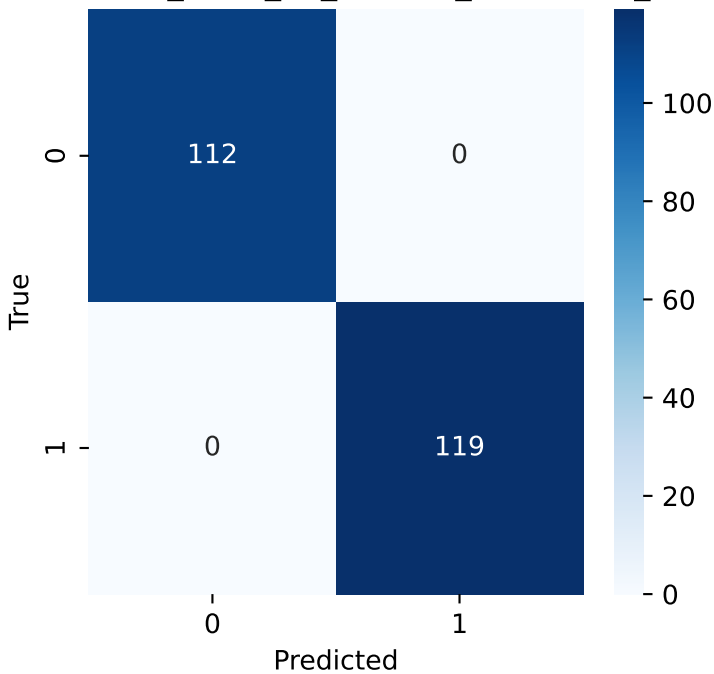
n Matrix: 07_2257_mi_session_20250626_23012



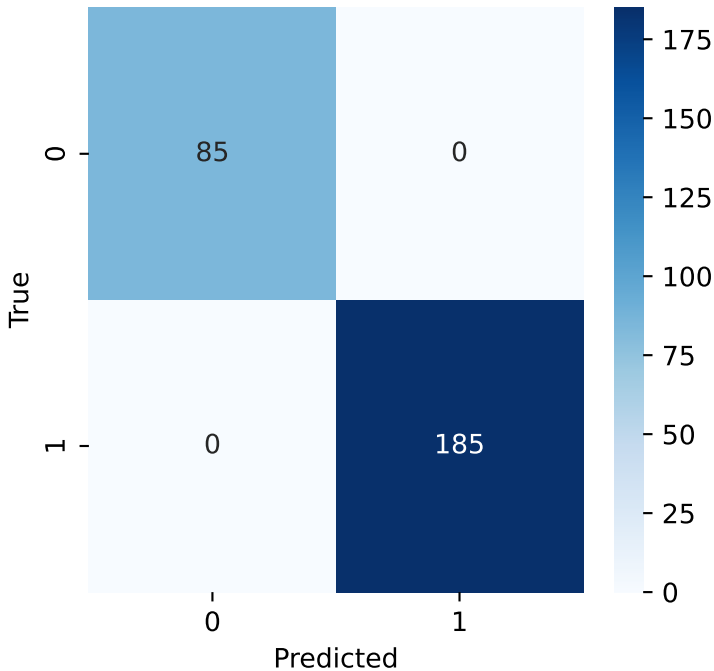
n Matrix: 08_2312_mi_session_20250626_23193



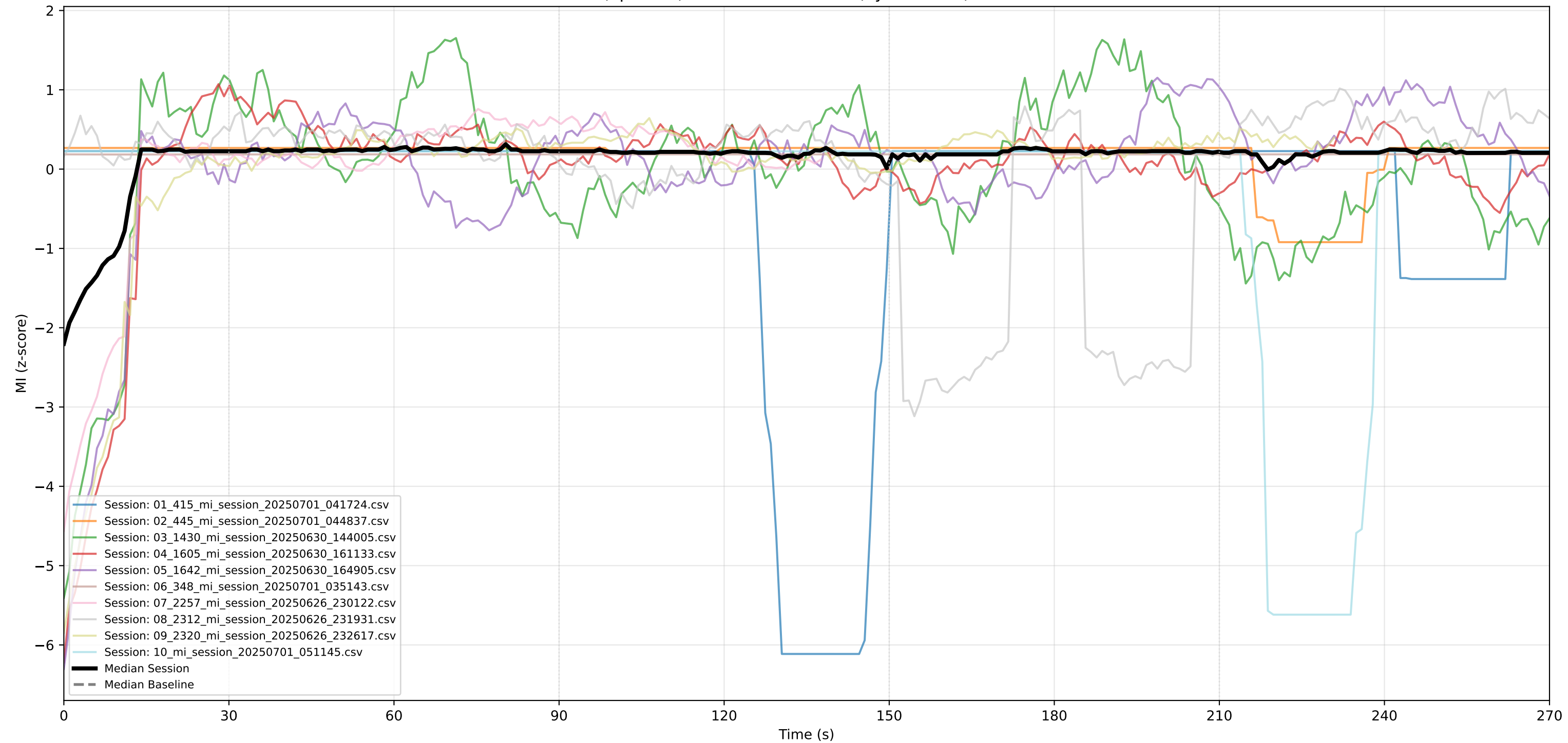
n Matrix: 09_2320_mi_session_20250626_23261



Confusion Matrix: 10_mi_session_20250701_051145.c



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 270s Duration



MI Advanced Report

Generated: 20250717_172803

MI Advanced Report - Statistical Summary (Experimental Protocol: 30s Calibration + 4x60s Colors, Sampling Rate: 1.000 Hz)

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 100 samples were included (100s minimum at 1Hz).
- * Individual and median MI curves were plotted with 20-second moving average (see Figure 8.1).
- * Experimental protocol duration: 270 seconds with 270 samples
- * Protocol structure: 30s calibration + 60s color1 + 60s color2 + 60s color3 + 60s color4
- * Calculated sampling rate: 1.000 Hz
- * RM-ANOVA across experimental epochs (Calibration 0-30s, Color1 30-90s, Color2 90-150s, Color3 150-210s, Color4 210-270s):

Anova

```
=====
F Value Num DF Den DF Pr > F
-----
Epoch      nan 4.0000 36.0000      nan
=====
```

* Post-hoc pairwise comparisons (Tukey HSD):

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1      group2      meandiff p-adj lower upper reject
-----
Calibration (0-30s) Color 1 (30-90s) 0.8818 nan nan nan False
Calibration (0-30s) Color 2 (90-150s) 0.5833 nan nan nan False
Calibration (0-30s) Color 3 (150-210s) nan nan nan nan False
Calibration (0-30s) Color 4 (210-270s) nan nan nan nan False
Color 1 (30-90s) Color 2 (90-150s) -0.2985 nan nan nan False
Color 1 (30-90s) Color 3 (150-210s) nan nan nan nan False
Color 1 (30-90s) Color 4 (210-270s) nan nan nan nan False
Color 2 (90-150s) Color 3 (150-210s) nan nan nan nan False
Color 2 (90-150s) Color 4 (210-270s) nan nan nan nan False
Color 3 (150-210s) Color 4 (210-270s) nan nan nan nan False
=====
```

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at sample times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- * AUC (median): 38.47; mean $\pm$ SD (individual): nan $\pm$ nan (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = -0.127, CI = [-0.090, 0.088].
- * Change point detection (Pelt algorithm): 1 change points at sample indices [10] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering (k=2): cluster averages {'cluster 0': {0: 0.21234298244696365, 1: 0.23686006515309224, 2: 0.27238033685319873, 3: 0.3155972563472895, 4: 0.2687212534691555, 5: 0.28940656857883396, 6: 0.26440439750292655, 7: 0.212277497480634, 8: 0.202564358212379, 9: 0.18948523355423436, 10: 0.2155525478004107, 11: 0.20381253503640884, 12: 0.20660496104553533, 13: 0.2497886025555009, 14: 0.24414648511025344, 15: 0.27116368609682795, 16: 0.27004670913089773, 17: 0.30008605498920166, 18: 0.2786923141436377, 19: 0.2639386347278259, 20: 0.2540141616463714, 21: 0.2382688715183316, 22: 0.21415172686611764, 23: 0.18458541912739737, 24: 0.2107551816130088, 25: 0.19319944218960514, 26: 0.21290001792343133, 27: 0.2584059984179886, 28: 0.27112581074877273, 29: 0.2882248643564109, 30: 0.27703637747546905, 31: 0.31472470631434585, 32: 0.32526605214902027, 33: 0.24828603583851563, 34: 0.251273902595523, 35: 0.26769792292360234, 36: 0.28330987490628334, 37: 0.2847141587279392, 38: 0.26382310004846976, 39: 0.2788295257636867, 40: 0.28702604234324103, 41: 0.2779708979710612, 42: 0.23827645566741537, 43: 0.24934487998352833, 44: 0.2696062904178951, 45: 0.28004057621159417, 46: 0.26277161179226216, 47: 0.25868998351501615, 48: 0.2780456347343062, 49: 0.2720225466576086, 50: 0.2791602690266826, 51: 0.26314304110005227, 52: 0.24995992743211443, 53: 0.29093231214990045, 54: 0.28645868434740956, 55: 0.2650694564065574, 56: